A PROJECT TITLE

ON

Insider Threat Detection Using Elastic Net Regularized Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF : COMPUTER SCIENCE & INFORMATION TECHNOLOGY

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1. ABSTRACT:

Insider threats pose a significant cybersecurity risk, as they originate from authorized users who already possess legitimate access to an organization's systems and sensitive data. This project proposes a Machine Learning-based Insider Threat Detection System that analyzes user activity data — including login patterns, failed login attempts, after-hours access, file and USB usage, downloads, and cloud uploads — to identify behavioral anomalies indicative of potential insider threats.

The collected data is preprocessed through cleaning, feature selection, encoding, and normalization to ensure reliable model performance. For classification, the system employs Elastic Net Regularized Logistic Regression, which combines L1 and L2 penalties to select relevant behavioral features, reduce overfitting, and handle correlated variables effectively. The model is trained and evaluated using precision, recall, F1-score, and AUC-ROC to ensure reliable detection while minimizing false positives.

Based on the analyzed behavior, users are assigned a risk score reflecting their likelihood of insider threat involvement. An administrative dashboard visualizes user activity, risk levels, and the key behavioral factors driving each classification, giving security teams interpretable insight into flagged cases. By enabling early identification of anomalous activity, the proposed system supports proactive investigation and helps organizations mitigate insider threats before they result in data loss or security breaches.

2. INTRODUCTION:

Organizations today rely heavily on digital infrastructure, granting employees, contractors, and partners access to sensitive systems and data to carry out their roles. While external cyberattacks often dominate security discussions, a growing share of data breaches originate from within the organization itself. Insider threats — whether driven by malicious intent, negligence, or compromised credentials — are particularly dangerous because the individuals involved already have authorized access, allowing suspicious activity to blend in with normal operations and evade traditional perimeter-based security tools.

Conventional security systems, such as firewalls and intrusion detection systems, are largely designed to detect external attackers and often fail to flag anomalies in the behavior of legitimate users. Detecting insider threats therefore requires a different approach: one that continuously monitors user behavior patterns and identifies deviations that may signal malicious or risky activity, such as unusual login times, repeated failed access attempts, abnormal file transfers, or unauthorized use of external storage devices.

This project addresses this gap by proposing a Machine Learning-based Insider Threat Detection System. By applying Elastic Net Regularized Logistic Regression to behavioral and access-log data, the system identifies the most relevant indicators of risk while remaining resistant to overfitting and correlated features. The resulting model generates interpretable risk scores and visual insights through an administrative dashboard, enabling security teams to move from reactive incident response to proactive threat identification — addressing insider risks before they escalate into data loss or security breaches.

3. SOFTWARE REQUIREMENTS:

Requirement	Specification
Operating System	Windows 10/11 or Linux
Programming Language	Python 3.9 or above
IDE	Google Colab
Machine Learning Library	Scikit-learn
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Machine Learning Model	Elastic Net Regularized Logistic Regression
Frontend Technologies	HTML, CSS, JavaScript
Database	MongoDB
Dataset Format	CSV
Model Evaluation	Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC-AUC
Web Browser	Google Chrome / Microsoft Edge
Version Control	Git / GitHub

Signature of the Guide **Course Instructor**